TITLE:Design a guest network isolated from corporate network. Use 1 router with VLANs for separation. Configure DHCP for both networks. Setup DNS server. Configure static routes to control access. Add a web server accessible from both networks.

OBJECTIVES:

- To design a guest network that is logically isolated from the corporate network using VLANs on a single router.
- To implement separate DHCP pools for the corporate and guest networks.
- To set up a DNS server to resolve domain names to IP addresses.
- To configure static routes to control and restrict access between the two networks.
- To deploy a web server accessible from both the corporate and guest networks.
- To verify network isolation and connectivity using ping and browser tests in Cisco Packet Tracer.

THEORY:

1. VLANs (Virtual Local Area Networks)

A VLAN is a logical grouping of network devices that communicate as if they were on the same physical segment, regardless of their physical location. VLANs are configured on a Layer 2 switch and enable network segmentation, improved security, and reduced broadcast domains. Devices in different VLANs cannot communicate without a Layer 3 device. In this lab, Router-on-a-Stick is used: a single physical router interface is divided into subinterfaces, each tagged with 802.1Q encapsulation for the respective VLAN. This allows the router to route between VLANs while maintaining logical separation.

2. DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns IP addresses and network configuration parameters (subnet mask, default gateway, DNS server) to devices. A DHCP server maintains pools of IP addresses and leases them to clients. In this lab, separate DHCP pools are configured for VLAN 10 (Corporate) and VLAN 20 (Guest), ensuring each group of devices receives appropriate network configuration without manual assignment.

3. DNS (Domain Name System)

DNS is a hierarchical naming system that translates human-readable domain names (such as www.company.com) into IP addresses. A local DNS server is configured to resolve internal domain names, allowing users to access the web server by name rather than IP address. Both corporate and guest clients point to the same DNS server, while access control is enforced via static routes.

4. Static Routing

Static routes are manually configured routes that explicitly define the path for network traffic. Unlike dynamic routing protocols such as RIP or OSPF, static routes do not change automatically. In this lab, static routes are used to control access between the corporate and guest VLANs, allowing both segments to reach the web server while preventing direct communication between the two VLANs for security isolation.

5. Web Server

A web server hosts web pages and responds to HTTP requests from clients. In this topology, a centrally located web server is placed on a dedicated VLAN (VLAN 30 — Server VLAN) and is reachable from both the corporate and guest networks via static routes. This simulates a shared resource scenario where a company provides internet-accessible services while still maintaining network segmentation.

NETWORK TOPOLOGY:

The network topology uses a single Cisco 2911 router connected to a Cisco 2960T-24 Layer 2 switch. The switch carries three VLANs: VLAN 10 for Corporate PCs, VLAN 20 for Guest devices, and VLAN 30 for the Server. The router handles inter-VLAN routing via the Router-on-a-Stick method using 802.1Q-tagged subinterfaces, DHCP for both VLANs, and static routing to control access.

- Router0 (Cisco 2911) — Handles Inter-VLAN routing, DHCP for both VLANs, static routes, and DNS forwarding.
- Switch0 (Cisco 2960T-24) — Layer 2 switch with VLANs 10, 20, and 30 configured; Fa0/24 to Router0 as trunk.
- PC0, PC1 — Corporate clients in VLAN 10, receive DHCP from 192.168.10.0/24.
- PC2, PC3 — Guest clients in VLAN 20, receive DHCP from 192.168.20.0/24.
- Server0 ,server1— Web server and DNS server with static IP in VLAN 30 (192.168.30.10 ,192.168.30.11).

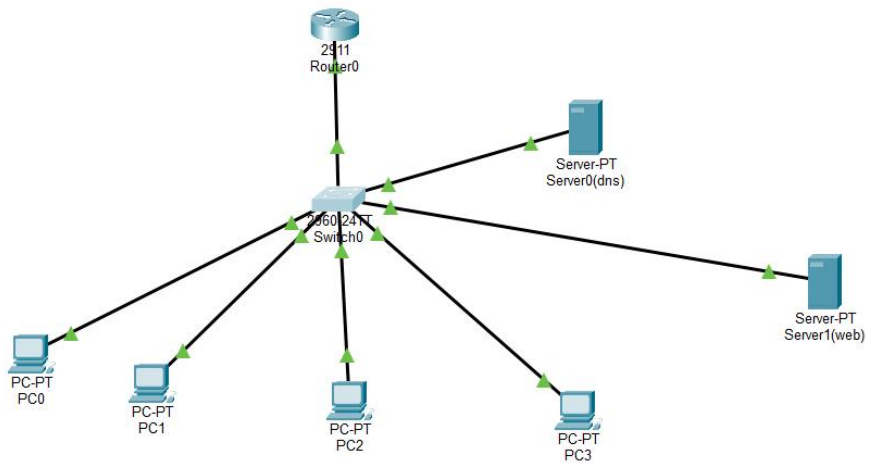

Fig: Network Topology

CONFIGURATION TABLE:

Device IP Address Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	Gi0/0.10	192.168.10.1	255.255.255.0	N/A
Router0	Gi0/0.20	192.168.20.1	255.255.255.0	N/A
Router0	Gi0/0.30	192.168.30.1	255.255.255.0	N/A
Server0	Fa0	192.168.30.10	255.255.255.0	192.168.30.1
Server1	Fa0	192.162.30.20	255.255.255.0	192.168.30.1
PC0	Fa0	DHCP (192.168.10.12)	255.255.255.0	192.168.10.1
PC1	Fa0	DHCP (192.168.10.11)	255.255.255.0	192.168.10.1
PC2	Fa0	DHCP (192.168.20.12)	255.255.255.0	192.168.20.1
PC3	Fa0	DHCP (192.168.20.11)	255.255.255.0	192.168.20.1

VLAN Assignment Table:

VLAN ID	VLAN Name	Switch Ports	Devices
10	Corporate	Fa0/1, Fa0/2	PC0, PC1
20	Guest	Fa0/3, Fa0/4	PC2, PC3
30	Server	Fa0/5,Fa0/6	Server0(DNS),Server1(Web)
Trunk	All VLANs	Fa0/24	Router0 (Gi0/0)

DHCP Pool Configuration:

Pool Name	Network	Gateway	DNS Server	Clients
CORPORATE_POOL	192.168.10.0/24	192.168.10.1	192.168.30.10	PC0, PC1
GUEST_POOL	192.168.20.0/24	192.168.20.1	192.168.30.10	PC2, PC3

Static Routing Configuration:

Device	Destination Network	Next-Hop / Interface	Purpose
Router0	192.168.30.0/24	Gi0/0.30 (directly connected)	Route to Server VLAN
Router0	192.168.10.0/24	Gi0/0.10 (directly connected)	Route to Corporate VLAN
Router0	192.168.20.0/24	Gi0/0.20 (directly connected)	Route to Guest VLAN

PROCEDURE:

The following steps were performed to configure the network topology in Cisco Packet Tracer:

Step 1: Configure VLANs on Switch0

VLANs were created on the switch and access ports were assigned to the respective VLANs. The uplink to Router0 was configured as a trunk port to carry all VLANs.

```
Switch>enable
Switch#configure terminal
Switch(config)#vlan 10
Switch(config-vlan)#name Corporate
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Guest
```

```
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name Server
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#end
Switch#write memory
```

Step 2: Configure Router0 Subinterfaces (Router-on-a-Stick)

The physical interface Gi0/0 was enabled without an IP address. Subinterfaces were created for each VLAN with 802.1Q encapsulation to enable inter-VLAN routing.

```
Router0>enable
Router0#configure terminal
Router0(config)#interface gigabitEthernet 0/0
```

```
Router0(config-if)#no ip address
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface gigabitEthernet 0/0.10
Router0(config-subif)#encapsulation dot1Q 10
Router0(config-subif)#ip address 192.168.10.1 255.255.255.0
Router0(config-subif)#no shutdown
Router0(config-subif)#exit
Router0(config)#interface gigabitEthernet 0/0.20
Router0(config-subif)#encapsulation dot1Q 20
Router0(config-subif)#ip address 192.168.20.1 255.255.255.0
Router0(config-subif)#no shutdown
Router0(config-subif)#exit
Router0(config)#interface gigabitEthernet 0/0.30
Router0(config-subif)#encapsulation dot1Q 30
Router0(config-subif)#ip address 192.168.30.1 255.255.255.0
Router0(config-subif)#no shutdown
Router0(config-subif)#exit
Router0(config)#end
Router0#write memory
```

Step 3: Configure DHCP Pools on Router0

Separate DHCP pools were configured for the Corporate (VLAN 10) and Guest (VLAN 20) networks. Excluded addresses were set to reserve the first 10 IPs for static assignments. Both pools point to Server0 as the DNS server.

```
Router0#configure terminal
Router0(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router0(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router0(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.10
Router0(config)#ip dhcp pool CORPORATE
Router0(dhcp-config)#network 192.168.10.0 255.255.255.0
Router0(dhcp-config)#default-router 192.168.10.1
Router0(dhcp-config)#dns-server 192.168.30.10
Router0(dhcp-config)#exit
Router0(config)#ip dhcp pool GUEST
Router0(dhcp-config)#network 192.168.20.0 255.255.255.0
Router0(dhcp-config)#default-router 192.168.20.1
Router0(dhcp-config)#dns-server 192.168.30.10
Router0(dhcp-config)#exit
Router0(config)#end
Router0#write memory
```

Step 4: Configure DNS Server (Server0)

Server0 was assigned a static IP address. The DNS service was enabled and DNS A records were added to resolve domain names to Server0's IP address. HTTP service was also enabled to make Server1 act as a web server.

IP Configuration (Desktop > IP Configuration > Static):

Field	Value
IP Address	192.168.30.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.30.1
DNS Server	192.168.30.10

DNS Records added (Services > DNS > ON):

Name	Record Type	IP Address
www.company.com	A Record	192.168.30.20

Step 5: Configure Static Routes to Control Access

Static routes are used to control which networks can reach the web server. The router's directly connected routes already handle intra-VLAN routing. To prevent the guest VLAN from reaching the corporate VLAN directly, access control is enforced by not advertising corporate routes to the guest subnet. Both VLANs can reach the Server VLAN (192.168.30.0/24) via the router.

Since all three VLANs are directly connected to Router0 via subinterfaces, the routing table already contains all necessary routes. To further isolate the Guest network from the Corporate network, an access control list (ACL) is applied on the router:

```
Router0#configure terminal
Router0(config)#ip access-list extended GUEST_ISOLATION
Router0(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Router0(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Router0(config-ext-nacl)#permit ip any any
Router0(config-ext-nacl)#exit
Router0(config)#interface gigabitEthernet 0/0.20
Router0(config-subif)#ip access-group GUEST_ISOLATION in
Router0(config-subif)#exit
Router0(config)#interface gigabitEthernet 0/0.10
Router0(config-subif)#ip access-group GUEST_ISOLATION in
```

```
Router0(config-subif)#exit
Router0(config)#end
Router0#write memory
```

RESULT:

Routing Table Verification (Router0):

```
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0.30

Router#
```

DHCP Binding Verification (Router0):

```
Router#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.11    0004.9AAA.4DDB           --                     Automatic
192.168.10.12    0001.C919.CA1A           --                     Automatic
192.168.20.11    0002.4A81.5D3D           --                     Automatic
192.168.20.12    0060.3E4D.A45C           --                     Automatic
Router#
```

VLAN Verification (Switch0):

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Gig0/1, Gig0/2
10   CORPORATE                        active    Fa0/1, Fa0/2
20   GUEST                            active    Fa0/3, Fa0/4
30   SERVER                           active    Fa0/5, Fa0/6
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Connectivity Test Results

Source	Destination	IP Address	Result
PC0 (Corporate VLAN 10)	Server1 Web Server	192.168.30.20	Success
PC2 (Guest VLAN 20)	Server1 Web Server	192.168.30.20	Success
PC0 (Corporate VLAN 10)	PC3 (Guest VLAN 20)	192.168.20.11	Blocked (ACL)
PC2 (Guest VLAN 20)	PC1 (Corporate VLAN 10)	192.168.10.11	Blocked (ACL)
PC0 (Corporate)	DNS Resolution (www.company.com)	192.168.30.10	Success
PC2 (Guest)	DNS Resolution (www.company.com)	192.168.30.10	Success
PC0 (Corporate)	HTTP to www.company.com	Browser test	Success
PC2 (Guest)	HTTP to www.company.com	Browser test	Success

1.Ping from (corporate Vlan10) to (server1 web server):

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.20

Pinging 192.168.30.20 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.20: bytes=32 time<1ms TTL=127
Reply from 192.168.30.20: bytes=32 time<1ms TTL=127
Reply from 192.168.30.20: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.20:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

2.Ping from (Guest Vlan20) to (server1 web server):

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.20

Pinging 192.168.30.20 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.20: bytes=32 time<1ms TTL=127
Reply from 192.168.30.20: bytes=32 time<1ms TTL=127
Reply from 192.168.30.20: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.20:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

3.Ping from (Guest Vlan20) to (corporate Vlan10) :

```
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

4.Ping from (corporate Vlan10) to (server0 DNS):

```
C:\> ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

5.Ping from (Guest Vlan20) to (server0 DNS):

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

6. Browser test from (corporate Vlan10) :

DISCUSSION:

In this experiment, a guest network was successfully designed and isolated from a corporate network using VLANs on a single Cisco 2911 router in Cisco Packet Tracer. The wired segment was divided into three logical networks: Corporate (VLAN 10), Guest (VLAN 20), and Server (VLAN 30). Inter-VLAN routing was achieved using the Router-on-a-Stick technique, where subinterfaces on Gi0/0 with 802.1Q encapsulation served as the default gateways for each VLAN.

A critical aspect of the lab was ensuring proper isolation between the corporate and guest networks while maintaining shared access to the web server. This was accomplished by applying an extended ACL (GUEST_ISOLATION) on both VLAN 10 and VLAN 20 subinterfaces. The ACL explicitly denied traffic between the corporate and guest subnets while allowing all other traffic, including traffic to the Server VLAN. This allowed both groups to access the web server and DNS service without being able to directly communicate with each other.

The DHCP configuration was straightforward, with separate pools (CORPORATE and GUEST) for each VLAN. Both pools were configured to point to Server0 (192.168.30.2) as the DNS server and Server1(192.168.30.3) as the web server. The DNS server was configured with A records mapping www.company.com static IP address. HTTP service was verified by opening a web browser on both corporate and guest PCs and navigating to the domain name, confirming successful DNS resolution and web server access.

The primary advantage of this topology is its simplicity: a single router manages routing, DHCP, and access control for the entire network. This is suitable for small to medium-sized deployments where a dedicated firewall is not available. In enterprise environments, more robust solutions such as dedicated firewalls or Layer 3 switches with proper ACLs would typically be used to enforce stricter guest isolation policies.

CONCLUSION:

The guest network isolation topology was successfully implemented in Cisco Packet Tracer. The corporate and guest networks were logically separated using VLANs on a single router, with DHCP automatically assigning IP addresses to clients in both segments. A dedicated DNS server resolved internal domain names, and a web server hosted on the Server VLAN was made accessible to both corporate and guest users. Static routes combined with ACL rules effectively enforced network isolation, preventing direct communication between the corporate and guest VLANs while allowing both to reach shared services. This experiment demonstrated a practical and scalable approach to guest network design using fundamental networking technologies including VLANs, DHCP, DNS, and access control lists.